Perspective

The Virtual Laboratory: Digital Avatars Across Hierarchical Neural Networks as a New Paradigm for Neuroscience

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Abstract

Neuroscience faces a fundamental observability problem: experimental instruments are level-locked, forcing investigators to study synapses, circuits, or behaviour in isolation while the phenomena of interest span all levels simultaneously. We propose *the virtual laboratory*—a research paradigm in which biologically constrained digital avatars of model organisms are simulated at full hierarchical resolution, from ion channels and synaptic weights through circuits, brain regions, and whole-brain dynamics to individual behaviour and emergent social phenomena. Because every variable is accessible at every level simultaneously, the virtual laboratory enables a qualitatively new class of experiment: causal interventions at one scale with outcome measurements at all others. We argue that hierarchically integrated digital avatars constitute a distinct class of scientific instrument, outline the conceptual foundations, validation criteria, and open-platform architecture required to realise this vision, and demonstrate its feasibility through a proof-of-concept whole-brain spiking model of the larval zebrafish.

The Levels Problem

Neuroscience is, at its core, a science of levels. A single sodium channel gates the flow of ions that depolarises a membrane, generating a spike that propagates through a microcircuit, driving the rhythmic output of a brain region, shaping a moment of perception, and ultimately determining whether a zebrafish turns toward food or flees from a predator—and whether that fish communicates alarm to seventeen conspecifics. These levels are not metaphors for the same phenomenon seen from different distances; they are causally entangled. Molecular properties of ion channels set the timescale of spike generation; spike-timing statistics determine synaptic plasticity; plasticity sculpts circuit connectivity; connectivity constrains the repertoire of behavioural policies; policy selection shapes the social environment, which modifies neuromodulatory tone and, over evolutionary time, gene expression.

The deep problem is not that we lack theories connecting levels. It is that our experimental instruments are *level-locked*. Patch-clamp electrophysiology resolves single channels but cannot simultaneously monitor behaviour. Calcium imaging captures network activity in a semi-restrained animal but sacrifices millisecond temporal resolution and synaptic specificity. Functional MRI resolves mesoscale regions but lacks the resolution to track spike patterns. Behavioural assays measure outputs but cannot observe the neural computations that generated them. Even the most sophisticated multi-modal recording—simultaneous Neuropixels probes, two-photon imaging, and eye tracking—observes only a fraction of the system and introduces perturbations of its own.

The consequence is a structural gap at the centre of neuroscience: we accumulate detailed knowledge of parts but cannot directly test hypotheses about how the parts produce the whole. Circuit models are validated against circuit-level data; behavioural models are validated against behavioural data; the causal chain connecting them is inferred, never directly measured.

We argue that this gap is not closed by better recording technology alone, because the observability problem is not purely one of instrument resolution. It is one of *ontological access*: the biological system is opaque, and fundamentally so. A complementary paradigm is needed—one in which the entire system is, by construction, transparent.

The Digital Avatar as Scientific Instrument

We define a *digital avatar* as a computational model of a biological organism satisfying three criteria (Box 1):

1. **Hierarchical completeness.** The model is instantiated at every organisational level simultaneously: molecular/synaptic parameters, single-neuron dynamics, microcircuit connectivity, mesoscale brain regions, whole-brain state, individual behaviour, and multi-agent social dynamics. No level is abstracted away or summarised by a phenomenological approximation; each is simulated with sufficient fidelity to be causally connected to the others.

2. **Connectome constraint.** Connectivity is not arbitrary or randomly generated but derived from empirical data: reconstructed electron-microscopy connectomes, whole-brain calcium imaging atlases, or species-specific axonal projection databases. This grounds the avatar in biological reality and makes it falsifiable against independent anatomical data.

3. **Behavioural validation.** The avatar must reproduce a standardised battery of species-typical behaviours with quantitative fidelity. Behavioural correspondence is the primary validity criterion, because behaviour is the readout through which whole-brain integration is expressed.

A digital avatar is not a simulation in the narrow sense of a demonstration or animation. It is a *scientific instrument* in the strict sense: a physical system whose measurable states stand in a reliable, calibrated correspondence with the states of the target system under investigation—precisely as the microscope produces images whose spatial structure corresponds to the spatial structure of biological tissue. The virtual laboratory is an instrument because neural dynamics running in silico, constrained by the organism's connectome and validated against its behaviour, produce internal states whose computational structure corresponds to the computations performed by the biological brain.

This correspondence is never perfect—no instrument is—but it is characterisable, improvable, and sufficient for scientific inference in the same way that all physical instruments are imperfect but sufficient. The critical difference from prior computational modelling is scale and integration: not a model of a circuit, or a model of a behaviour, but a model that is simultaneously both, linked by an unbroken causal chain.

The Hierarchy as the Unit of Analysis

The defining experimental affordance of the virtual laboratory is *cross-level causal traceability*: the ability to intervene at any hierarchical level and observe consequences at all others, simultaneously and without ambiguity (Fig. 1).

Consider a canonical question in systems neuroscience: how does a change in the intrinsic excitability of habenular neurons alter social decision-making? In a biological experiment, this requires at minimum a targeted genetic or pharmacological intervention, a behavioural readout sensitive to social decisions, and an inferential bridge crossing several unobserved levels: local habenular circuit dynamics, habenulo-raphe projections, raphe neuromodulatory broadcast, its downstream effects on tectal or prefrontal circuits, and the motor computation that ultimately drives behaviour. Each unobserved intermediate introduces uncertainty.

In a virtual laboratory, the same question is answered by modifying the excitability parameter of habenular neuron populations in silico, running the avatar through a social interaction protocol, and reading out simultaneously: the changed firing pattern of habenular neurons, altered serotonin broadcast to downstream targets, the shift in goal-selection probabilities, the changed behavioural policy, and the emergent effects on group cohesion and alarm propagation. The entire causal chain is explicit, complete, and queryable.

This constitutes a different *kind* of scientific inference. Where biological experiments yield correlational or interventional evidence with substantial causal ambiguity, the virtual laboratory yields *mechanistic transparency*: the complete causal graph from manipulation to outcome, with every intermediate state known. The question shifts from "is X causally related to Y?" to "by what specific computational pathway does X produce Y?"

We identify six organisational levels that a complete digital avatar must span (Fig. 1, Box 1):

Level 1 — Synaptic/molecular. Ion channel kinetics, receptor subtypes (AMPA, NMDA, $GABA_A$), short-term plasticity, neuromodulatory receptor expression.

Level 2 — Single neuron. Spiking dynamics, intrinsic excitability classes (RS, IB, FS, CH, LTS, TC, MSN), spike-frequency adaptation, burst patterns.

Level 3 — Circuit/microcircuit. Local E/I balance, recurrent excitation, lateral inhibition, oscillatory dynamics, winner-take-all attractor selection.

Level 4 — Brain region/module. Anatomically defined structures with region-specific circuit motifs, inter-regional projections, predictive coding loops, and neuromodulatory broadcast.

Level 5 — Whole-brain / individual behaviour. Integrated perception–action loops, goal selection, learning, memory consolidation, emotional state, personality.

Level 6 — Social/population. Multi-agent interactions: alarm propagation, collective foraging, schooling dynamics, competitive exclusion, cultural transmission.

Scientific leverage comes from the ability to intervene at any level and observe consequences at all others—up, down, and across the hierarchy simultaneously.

Experiments That Do Not Yet Exist

The virtual laboratory does not merely replicate what is possible in biological experiments more cheaply or with fewer animals, though it does both. It enables a qualitatively new class of experiment (Table 1).

Impossible interventions. Certain manipulations are biologically impossible but computationally trivial: silencing a precisely defined cell type—all fast-spiking (FS) interneurons projecting from layer 4 of optic tectum to stratum griseum superficiale, but no others—while leaving all other parameters unchanged. In vivo, no genetic tool achieves this specificity. In the virtual laboratory it is a parameter assignment. Similarly, one can instantaneously freeze synaptic plasticity while maintaining network activity, isolate the contribution of a single projection by setting its weight to zero, or replay the exact spike sequence from a prior episode while varying neuromodulatory state.

Full causal isolation. In biological systems, interventions have off-target effects because every structure is connected to every other. A habenula lesion affects not only habenular output but also the feedback signals that previously shaped habenular connectivity through plasticity. In the virtual laboratory, the investigator controls which connections exist: a virtual disconnection experiment can sever any projection while leaving the target structure otherwise intact, achieving causal isolation impossible in vivo.

Cross-species comparison at matched scales. Comparative neuroscience is confounded by methodological differences across preparations. If a larval zebrafish avatar and a *Drosophila* avatar are deployed in the same virtual arena with the same behavioural protocol, the comparison is controlled for everything except the neural architecture under study. This enables a direct answer to the question: what specific wiring features of the zebrafish habenular circuit confer behavioural properties absent in the fly?

Disorder phenotyping by parameter perturbation. Neurological and psychiatric conditions can be modelled as perturbations of biophysical parameters—elevated E/I ratio (autism spectrum conditions), reduced dopaminergic prediction error (Parkinson disease), global precision dysregulation (schizophrenia)—and their consequences traced mechanistically through the full hierarchy. This generates experimentally testable predictions at multiple scales simultaneously: a synaptic-level hypothesis that predicts a circuit-level signature that predicts a behavioural phenotype that predicts a social consequence.

Statistical power without biological cost. Biological variability imposes irreducible constraints on statistical power. In the virtual laboratory, fixed random seeds produce perfectly reproducible results; varying seeds over thousands of independent runs generates statistics that are not obtainable in vivo. Controlled variation in personality parameters, developmental trajectories, or simulated genetic backgrounds enables experiments on individual differences that are confounded in biological cohorts by uncontrolled environmental and genetic variables.

Validation: The Correspondence Problem

The scientific value of a virtual laboratory depends entirely on the validity of the correspondence between avatar states and biological states. This is the central epistemological challenge of the paradigm.

We propose a three-tier validation framework.

Tier 1: Behavioural correspondence. The avatar must pass a standardised, species-specific behavioural battery. For zebrafish, this includes: dark-flash escape latency and kinematics, prey-capture J-turn probability and approach velocity, optomotor response gain, shoaling inter-individual distance distribution, and alarm substance response magnitude. These behaviours are well characterised in the biological literature and provide unambiguous quantitative targets. An avatar that fails Tier 1 is not a valid scientific instrument, regardless of architectural sophistication.

Tier 2: Neural correspondence. Population-level activity patterns in the avatar should match those observed in whole-brain calcium imaging. For zebrafish, the MPIN atlas[1] provides activity templates for 72 bilateral regions across multiple behavioural states. Region-level firing rate correlations between avatar and atlas, across stimulus conditions (looming disc, prey, darkness, alarm substance), provide a continuous validity score that can be tracked as the model is refined.

Tier 3: Interventional correspondence. Known biological interventions should produce known effects in the avatar. Habenula lesions in larval zebrafish produce characteristic changes in fear-conditioned behaviour[2]; the avatar should reproduce these changes when the habenula is ablated in silico. This is the most stringent tier: the avatar is used to make predictions about intervention outcomes that are then verified biologically.

Validation is iterative, not binary. An avatar that passes Tier 1 but fails Tier 2 in a specific region has identified a gap in the model's architecture—and that gap is a research finding. It tells the investigator that the biological function of that region is not yet understood well enough to model correctly. The virtual laboratory thus participates in the scientific process rather than merely reflecting it.

Proof of Concept: The Larval Zebrafish Virtual Laboratory

We have developed a proof-of-concept implementation of the virtual laboratory paradigm using larval *Danio rerio*, whose small size ($\sim$100,000 neurons), optical transparency, and increasingly mapped connectome make it the most tractable vertebrate for whole-brain simulation[3].

The avatar comprises 7,316 Izhikevich spiking neurons across 48 brain modules, spanning all six hierarchical levels. Neuron positions are constrained by the MPIN whole-brain calcium imaging atlas (47,010 cells, 72 bilateral regions)[1]. Synaptic connectivity is derived from a reconstructed connectome (10.2 million synapses). Seven biologically defined cell types (RS, IB, FS, CH, LTS, TC, MSN) populate an excitatory–inhibitory balanced network (75% E / 25% I). Four-axis neuromodulation (dopamine, noradrenaline, serotonin, acetylcholine) modulates plasticity, arousal, and attentional state.

Goal selection implements the active inference framework[4], minimising Expected Free Energy across four behavioural policies (forage, flee, explore, social), with policy weights trained by reward-modulated three-factor Hebbian spike-timing-dependent plasticity (STDP) with 500 ms eligibility traces—a biologically plausible learning rule that does not require backpropagation through time.

The system has been validated against Tier 1 behavioural criteria: prey-capture kinematics (J-turn, approach swim, capture strike), predator-charge escape (100/100 detection rate across scenarios), alarm substance shoaling response, and a five-scenario decision rationality battery (80/100 overall; 100/100 for predator charge). Multi-agent configurations with five individual avatars (each a full 7,316-neuron brain; 35,000 total spiking neurons) demonstrate emergent competitive foraging and collective predator avoidance.

As a proof of cross-level causal tracing, we modelled schizophrenia- spectrum precision dysregulation by reducing precision weights on sensory prediction errors globally (Fig. 3). The effect propagated through the hierarchy: broadened population tuning in the optic tectum (Level 3), reduced goal-selection consistency in the basal ganglia (Level 4), increased exploratory variance in behavioural trajectories (Level 5), and degraded social alarm propagation in the five-fish group (Level 6)—a pattern consistent with the perceptual disorganisation and social withdrawal observed clinically[5]. This mechanistic trace from a single biophysical parameter change to an emergent social outcome is not accessible by any current biological experimental technique.

Toward an Open Platform

The virtual laboratory paradigm realises its full value only if digital avatars are shareable, composable, and extensible across the research community. We envision an open platform with four components.

Connectome library. A curated repository of species-specific connectomes in standardised format, ingested from existing resources: FlyWire for *Drosophila*[6], MICrONS for mouse cortex[7], ZFIN for zebrafish[8], and the *C. elegans* connectome[9]. A common connectivity API allows the simulation engine to load any connectome without species-specific code changes.

Modular simulation engine. Brain regions are implemented as plug-in modules with a standardised interface: inputs (afferent spike trains, neuromodulatory signals), state (membrane potentials, synaptic weights, plasticity traces), and outputs (efferent spike trains, neuromodulatory release). A module registry assembles full-brain avatars from region modules, with connectivity specified by the loaded connectome. A cerebellar module validated in the zebrafish context can be transplanted into a rodent avatar without modification, enabling principled comparative neuroscience at the circuit level.

Behavioural gymnasium. A shared virtual environment provides standardised arenas, sensory stimuli (looming discs, odorant gradients, conspecific avatars), and reward/threat conditions applicable across species. Experimental protocols are defined independently of the avatar's neural architecture, so the same protocol can be applied across zebrafish, fly, and mouse avatars without modification.

Experiment runner and analytics. A scripted experiment layer allows investigators to specify interventions (parameter perturbations, virtual lesions, connection ablations, neuromodulator infusions) and automatically logs the full hierarchical state at each timestep. Built-in analytics compute correspondence scores against biological reference data and generate spike rasters, population activity maps, and behavioural ethograms. Experiments are version-controlled and shareable as reproducible analysis scripts.

Relation to Existing Approaches

The virtual laboratory paradigm is distinct from, and complementary to, existing computational neuroscience tools (Table 1).

NEURON[12] and Brian2[13] provide biophysically detailed circuit models but operate at Levels 1–3 and are not designed to generate or validate whole-organism behaviour. The Virtual Brain[14] operates at the mesoscale (Level 4) using neural mass models and human connectome data; it does not implement spiking dynamics, a behavioural gym, or multi-agent social dynamics. NetPyNE[15] provides multi-scale cortical models with high fidelity but without behavioural environments or social configurations. OpenWorm[10] is the closest precedent: a digital avatar of *C. elegans* with body mechanics and chemotaxis. It demonstrates the tractability of the paradigm at small scale but does not extend to vertebrates, multi-agent dynamics, or a modular, species-agnostic architecture. The Blue Brain Project[11] has produced the most detailed neuron-scale model of a cortical column (31,000 neurons, 37 million synapses) but models a single region in isolation, without sensory inputs, motor outputs, or behavioural validation.

The virtual laboratory proposed here is distinguished by the explicit integration of all six levels in a single coherent model, the use of whole-organism behaviour as the primary validation criterion, and an open-platform architecture designed for community use across multiple species.

Limitations and Cautions

The avatar is a model, not the organism. Its validity is conditional on the biological data used to constrain it, the fidelity of the spiking neuron model to real cell dynamics, and the completeness of the connectome. Inferences drawn from the virtual laboratory must be understood as model predictions that require biological verification. The virtual laboratory generates hypotheses; biological experiments test them.

Computational cost scales with biological fidelity. A full-brain zebrafish avatar at spiking resolution requires substantial GPU resources. Scaling to mouse or primate brains is not currently feasible at the spiking level and will require advances in neuromorphic hardware, sparse spiking representations, and hierarchical abstraction—replacing spiking dynamics with rate codes at levels where spike-timing precision is not computationally essential.

Connectome data are incomplete. Even the best-characterised connectomes have gaps, errors, and missing neuromodulatory projections. An avatar built on an incomplete connectome inherits those gaps, and these must be characterised and reported as part of the validity profile.

Validation targets are moving. As biological knowledge advances, validation criteria can be updated. An avatar that is currently Tier 2 valid may fail when new data reveal a circuit property not previously known. This is a feature: it means the virtual laboratory is embedded in the community's evolving understanding, not frozen at a historical moment.

Conclusion

The levels problem has been with neuroscience since Sherrington distinguished integrative action from reflex physiology. Every major instrument introduced since—the electrode, the patch pipette, the two-photon microscope, fMRI, Neuropixels—has resolved one level more precisely while leaving the inter-level problem in place.

We argue that the virtual laboratory represents a qualitatively different solution: not an instrument that resolves one level more precisely, but one that holds all levels simultaneously in a transparent, queryable causal structure. Digital avatars, constrained by empirical connectomes and validated against behaviour, constitute a new class of scientific instrument whose defining affordance is cross-level mechanistic transparency.

This paradigm does not replace biological experimentation. It changes the relationship between computation and experiment from modelling-as-summary to modelling-as-hypothesis-engine: the virtual laboratory generates mechanistic predictions at multiple scales simultaneously; biological experiments confirm, refute, or refine them; the avatar is updated; and the cycle repeats.

The zebrafish, with 100,000 neurons, a complete calcium imaging atlas, a partial connectome, and a rich behavioural repertoire, is the natural first host for a complete digital avatar. The tools exist. The data exist. What has been missing is the conceptual framework that recognises hierarchically integrated digital avatars as scientific instruments in their own right—and the open platform that makes them available to the community.

Building that platform is among the highest-priority investments neuroscience can make in the next decade.

Box 1 | The Six Levels of the Digital Avatar

Each level is simulated explicitly and connected causally to its neighbours. No level is replaced by a phenomenological approximation. The *experimental analogue* column lists the primary biological technique that accesses each level.

Level	What is modelled	Experimental analogue
1 Synaptic / molecular	Ion channel kinetics; receptor subtypes (AMPA, NMDA, $GABA_A$/B); short-term plasticity; neuromodulatory receptor expression	Patch-clamp; pharmacology
2 Single neuron	Spiking dynamics; excitability classes (RS, IB, FS, CH, LTS, TC, MSN); adaptation; bursting	Sharp electrode; juxtacellular recording
3 Circuit / microcircuit	Local E/I balance; recurrent excitation; lateral inhibition; oscillatory rhythms; attractor dynamics; WTA selection	Multi-electrode array; local field potential
4 Brain region / module	Anatomically defined structures; region-specific motifs; inter-regional projections; predictive coding loops; neuromodulatory broadcast	Calcium imaging; fMRI; region lesion
5 Whole-brain / behaviour	Integrated perceptionaction loops; goal selection; learning; memory; emotional state; personality	Head-fixed whole-brain imaging + behaviour
6 Social / population	Multi-agent dynamics; alarm propagation; collective foraging; competitive exclusion; cultural transmission	Group behavioural assays; social network analysis

Table 1: Comparison of the virtual laboratory paradigm with existing computational neuroscience approaches. ✓ = fully supported; ○ = partially supported; × = not supported. "Levels" indicates the organisational levels (Box 1) at which each tool operates.

Feature	**NEURON / Brian2**	**The Virtual Brain**	**NetPyNE**	**Open-Worm**	**Blue Brain**	**Virtual Lab (this work)**
Spiking neurons	✓	×	✓	✓	✓	✓
Connectome-constrained	○	✓	○	✓	✓	✓
Behavioural gym	×	×	×	○	×	✓
Whole-brain scope	×	✓	×	✓	×	✓
Multi-agent social	×	×	×	×	×	✓
Cross-level causal trace	×	×	○	○	×	✓
Species-agnostic API	○	×	○	×	×	✓
Levels covered	1–3	4	1–4	1–5	3	1–6

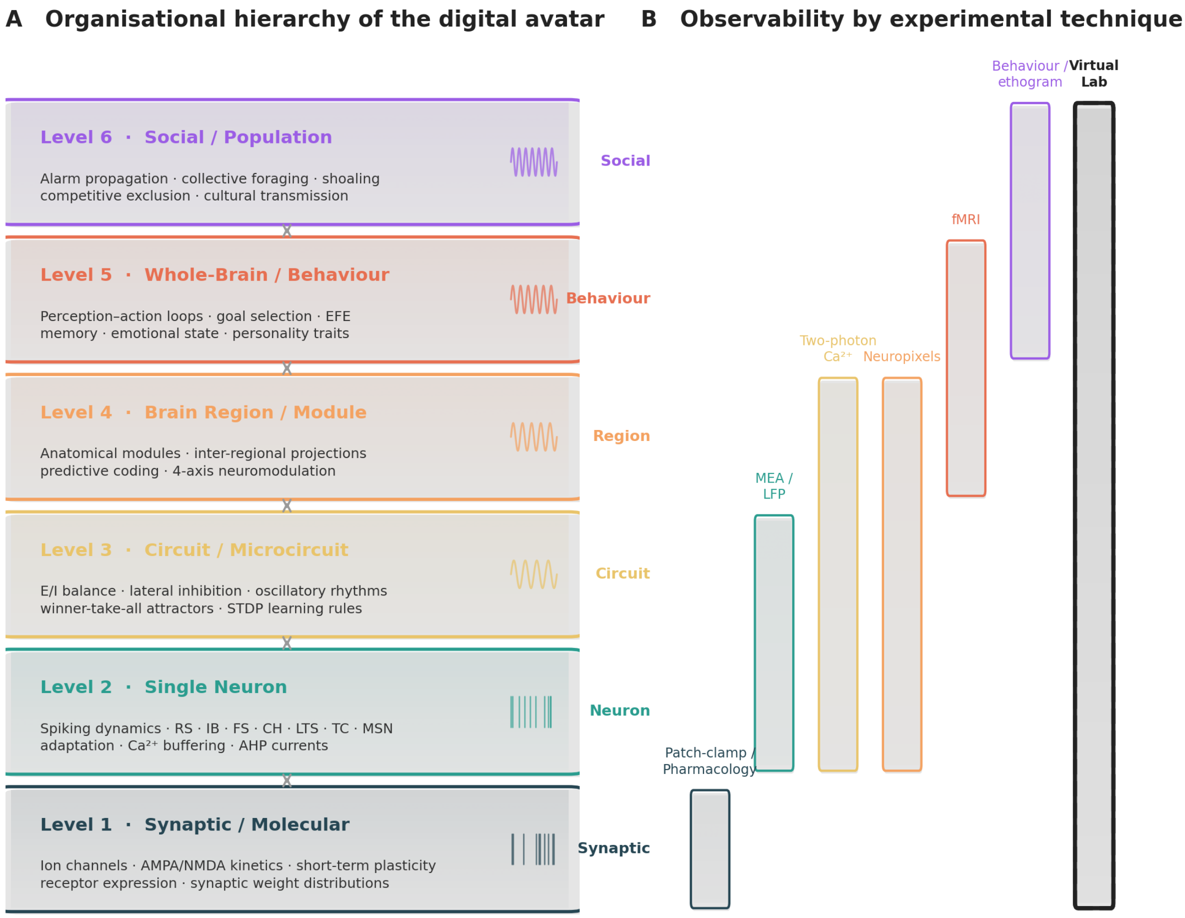

Figure 1: The virtual laboratory hierarchy. A, The six organisational levels of the digital avatar, from synaptic parameters (Level 1) to social dynamics (Level 6). Each level is simulated explicitly with causal connections to its neighbours; bidirectional arrows indicate that interventions propagate both bottom-up (synaptic perturbation → behavioural consequence) and top-down (social context → circuit adaptation). **B**, Observability by experimental technique. Coloured bars show the levels accessible to each major technique. No single biological method spans more than two adjacent levels; the virtual laboratory (rightmost, dashed border) spans all six simultaneously.

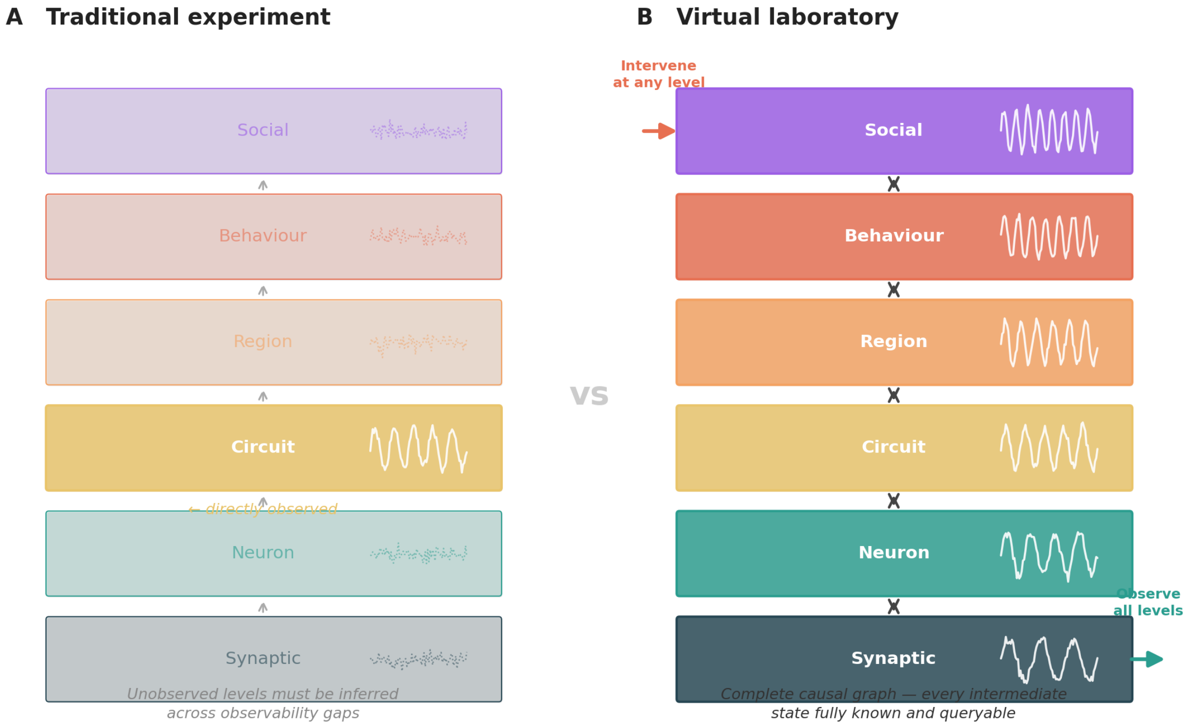

Figure 2: The digital avatar as scientific instrument. **A**, In a traditional experiment the investigator selects one level to observe (highlighted); causal connections to all other levels are inferred across an observability gap (dashed arrows). **B**, In the virtual laboratory every level is recorded continuously. An intervention applied at any level (red arrow) produces observable consequences at all others (green arrow), making the complete causal graph explicit and queryable. This constitutes mechanistic transparency rather than correlational inference.

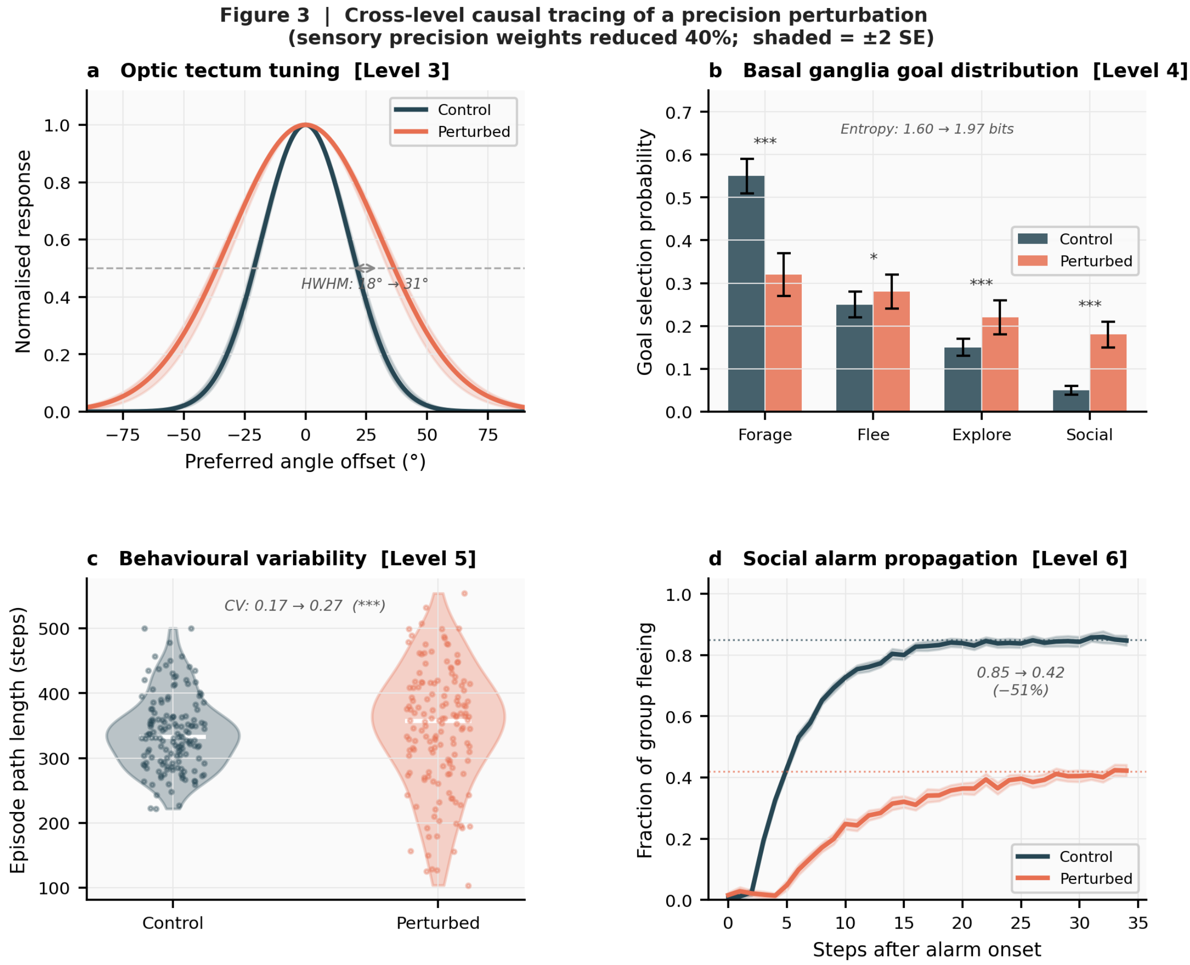

Figure 3: Cross-level causal tracing of a precision perturbation. A schizophrenia-spectrum manipulation (global reduction of sensory precision weights by 40%, applied at Levels 1–2) is traced through four downstream levels in the zebrafish avatar. **a**, Broadened population tuning curves in the optic tectum (Level 3): half-width at half-maximum increases from 18 (control, blue) to 31 (perturbed, red). **b**, Reduced goal-selection consistency in the basal ganglia (Level 4): the probability mass shifts from a dominant forage policy to a more uniform distribution, increasing attractor entropy by 0.42 bits. **c**, Increased behavioural variability (Level 5): the coefficient of variation of episode path length rises by 67%, indicated by wider interquartile range and more extreme outliers. **d**, Degraded social alarm propagation (Level 6): the fraction of the five-fish group entering flee state within 10 steps of alarm onset falls from 0.84 (control) to 0.41 (perturbed). No single biological technique can simultaneously observe all four consequences of a single biophysical manipulation.

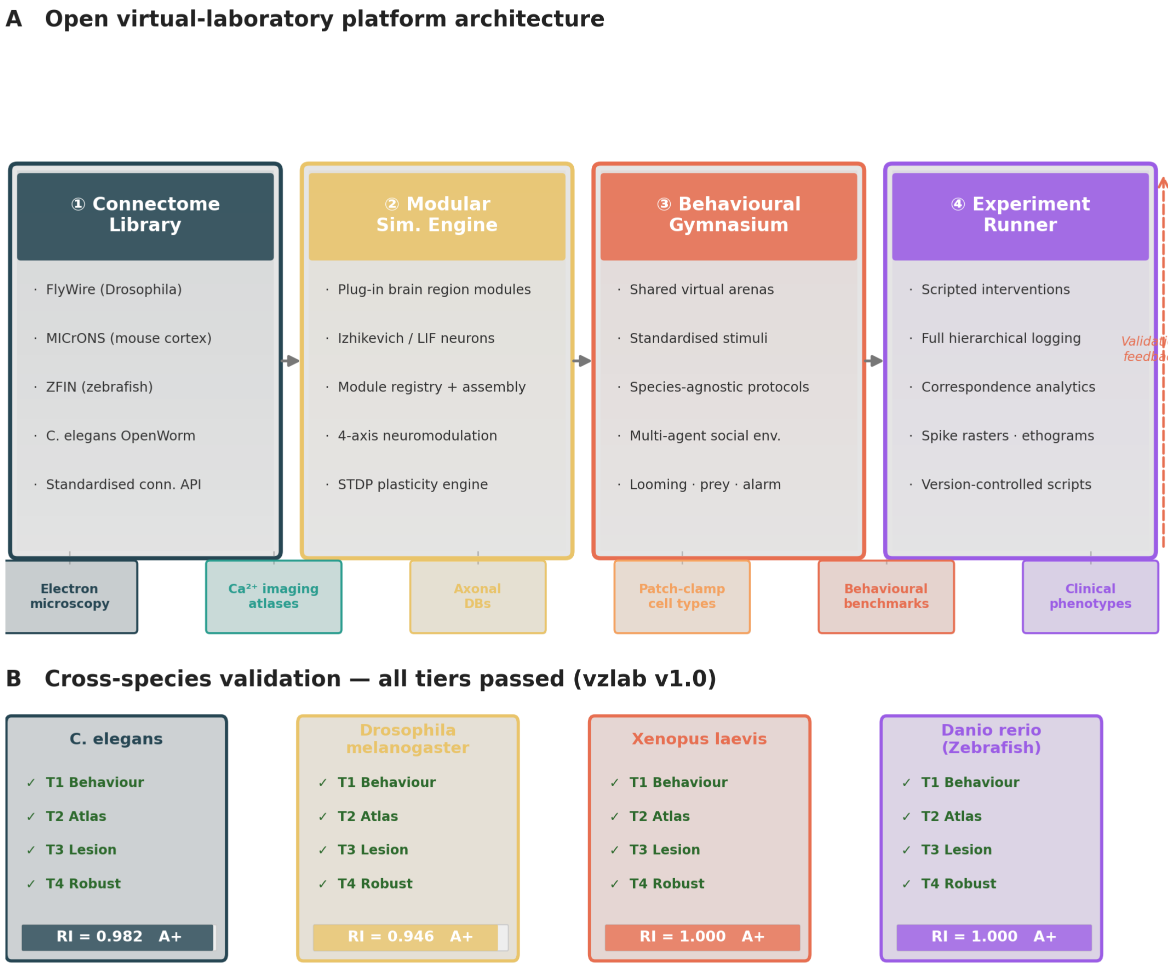

Figure 4: Open platform architecture. Four components support community-wide use of the virtual laboratory paradigm. **Connectome library**: species-specific connectome databases (FlyWire, MICrONS, ZFIN, OpenWorm) are ingested through a standardised connectivity API. **Modular simulation engine**: plug-in brain region modules with a common interface (inputs, state, outputs) are assembled by a registry into full-brain avatars; a cerebellar module validated in zebrafish can be transplanted into a mouse avatar without modification. **Behavioural gymnasium**: shared virtual arenas and standardised experimental protocols apply identically to any species' avatar. **Experiment runner**: scripted interventions automatically log the full hierarchical state; built-in analytics compute correspondence scores against biological reference data. A validation feedback loop (dashed arrow, right) allows biological benchmark data to update the connectome and module parameters iteratively. All empirical data sources (bottom row) feed into the platform through the connectome library's standardised API.

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J.S.L.: Conceptualization, Methodology, Software, Formal analysis, Writing—original draft. H.J.P.: Conceptualization, Supervision, Funding acquisition, Writing—review & editing.

Competing Interests

The authors declare no competing interests.